Research on Coordinated Adjustment Method of System Protection and Communication System under the Typical Fault Scenarios

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***Abstract*—With the rapid development of ultra-high voltage (UHV) inter-regional interconnected power grid and the wide application of a large number of power electronic devices, the security and stability characteristic of power grid is becoming more and more complex, and the risk of safe operation is increasing day by day. To further adapt to the power grid situation, in recent years, the State Grid Corporation of China has put forward the requirement of building a comprehensive security defense system for the large power grid, namely "system protection". System protection is an emergency control system based on wide area information collection, rapid fault diagnosis and isolation, multi-resource collaboration, and other functions. It covers many sites and the coupling relationship between device strategies is complex. At the same time, due to the limitation of control resources, the cross configuration between strategies is gradually increasing, and the influence range of local system anomalies is becoming larger and larger. Its complexity and hidden dangers lead to a significant increase in the risk of power grid operation. At present, there is little literature on how to reduce the risk of power grid safe and stable operation caused by the failure of the system protection itself through effective coordinated adjustment. In this paper, a coordinated adjustment method based on typical expected fault scenarios of system protection is proposed. For typical expected fault scenarios of system protection with unacceptable risks, certain measures are taken in advance to reduce the possible risks to the power system.**

Keywords—system protection, coordinated adjustment, typical fault scenarios.

I. INTRODUCTION

In a power system, the emergency load shedding and emergency generator tripping operation of system protection is an important measure to ensure the safe and stable operation of power grid [1]. However, with the enhancement of the complexity of system protection business and the limitation of controllable resources in the system, the coupling between various devices and functions of system protection becomes very close. The failure of information equipment or communication link in the system protection will lead to a significant decline in the service execution ability of the system protection and affect the security of the power system. Several blackouts occurred in India because the low-frequency action device of the power grid failed to act in time and effectively, which had a serious impact on the power grid. With the construction of AC / DC hybrid power grid in China and the wide access of new energy in the generation side, the power system has higher and higher requirements for system protection, and more sophisticated and complex system protection is required to meet the challenges of the power grid. However, in practice, the system protection hardware equipment, software or communication equipment are faced with various risks (the risks may come from man-made malicious damage or bad weather conditions, etc.), resulting in a significant decline in the ability of system protection. Therefore, it is particularly important to carry out a risk assessment on the possible fault scenarios of system protection, and it is more urgent for the power grid to take measures to reduce the risk of power system when the risk assessment is not up to standard.

At present, there are some identification and calculation methods for the actual executable strategy under the system protection expected fault scenarios [2], but there are still some problems in how to implement decision adjustment when the fault risk of system protection is found to be unacceptable. It is found that different adjustment measures of the communication system can reduce the risk of the power system to a certain extent [3]. Therefore, this paper proposes a coordinated adjustment method of emergency control strategy and

This work was supported by State Grid Corporation of China project (File NO. 5100-202040014A-0-0-00): Research on Key Technology of System Protection Sand Table Deduction System.

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communication system under the system protection typical expected fault scenarios.

II. Duty Strategy of System Protection in the Expected Failure Scenarios

A. System Protection Strategy Structure

In practical engineering applications, the strategy of system protection is stored in the control master station of system protection in a certain format [4]. Generally speaking, the standard structure of system protection strategy is generally divided into two parts: criterion constraints and execution action [2]. The criterion constraints include power system operation mode (system protection operation mode), operation power flow and power system fault location.

Operation mode indicates the operation mode of the power system at this time. The system protection needs to be matched with the operation mode of the power system by setting a pressing plate. Operation power flow depends on the output power of different generators. Fault location indicates the location of the fault. Execution action indicates the total amount of generator tripping and load shedding.

B. Duty Strategy of System Protection in the Expected Failure Scenarios

Due to the certain failure risk of system protection, it is necessary to identify the actual executable control strategy (duty strategy) under the expected failure scenarios of system protection [2]. When the system protection strategy is found to be invalid, the control strategy needs to be adjusted. The duty strategy indicates the amount of generator tripping and load shedding that the system can actually implement.

The actual executable strategy of system protection in the expected failure scenarios mainly depends on three factors: power system operation mode (system protection operation mode), system protection operation state in the expected failure scenarios and system protection operation pattern.

1) Power system operation mode

The operation mode of power system will determine the operation mode of system protection. Specifically, for different operation modes of power system, the control strategy of system protection needs to change with them. Under different operation modes, the system protection needs to consider different faults, and then there will be different strategies. Generally, the strategies under different operation modes will be obtained through simulation verification in advance [5]-[7].

2) Operation state of system protection under expected fault scenarios

Some strategies of system protection under the scenarios of expected failure may fail, so it is necessary to study the status of system protection under the scenarios of expected failure. Since there are many fault scenarios in the system protection, only the system protection status under typical fault scenarios is studied here.

A typical system protection structure is shown in Fig. 1 below. According to the following example, it shows how to obtain typical fault scenarios of system protection through scenario reduction and definition.

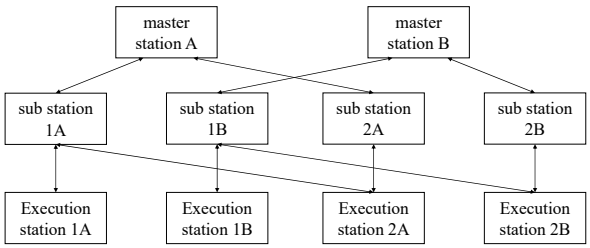


Fig. 1. The schematic diagram of system protection structure.

According to the schematic diagram of the system protection structure, the function logic function of the executive station 1 can be obtained as follows:

$$S_{P1} = MA \cdot L_{MA-S1A} \cdot S1A \cdot L_{S1A-E1A} \cdot E1A + MB \cdot L_{MB-S1B} \cdot S1B \cdot L_{S1B-E1B} \cdot E1B \quad (1)$$

S_{Pi} indicates the function status of each executive station; MA(Master Station A) indicates the status of the master station; L indicates the link status; SA(Sub Station A) indicates the status of the sub station; EA(Execution Station A) indicates the status of the executive station; A and B indicate that the system protection adopts dual configuration.

Since the system protection adopts a dual configuration, the N−1 fault will not have an impact, and the probability of N−3 and above faults is too low to be considered. Therefore, the N−2 fault scenarios of the system protection equipment are taken as the research scope. According to the arrangement and combination, it can be known that the executive station 1 has $C_{10}^{2} = 45$ fault scenarios, and then it is further obtained that the number of effective fault scenarios is $C_5^1 \cdot C_5^1 = 25$. The fault scenarios and the number of effective fault scenarios for other functions are shown in Table I.

After determining the effective fault scenarios by scenario reduction, determine the typical fault scenarios according to the definition. Then, the status of different functions of system protection under typical fault scenarios can be calculated according to the logic function. Definition of typical fault scenarios of system protection: under different fault scenarios of system protection, traverse the expected fault of power system, accumulate the sum of consequences caused by system protection fault, and take the fault scenarios whose fault consequences exceed the threshold as typical fault scenarios.

TABLE I. Number of Effective Fault Scenarios

Function	Number of fault scenarios	Number of effective fault scenarios
P1	45	25
P2	91	4

3) Operation pattern of system protection

The operation pattern of system protection indicates the relationship between two sets of system protection [8]. After obtaining the status of a set of system protection, the final duty strategy is obtained through strategy combination according to the operation pattern of system protection. If the system protection adopts single configuration, this step is not required. For the system protection with dual configuration, the final

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strategy can be obtained only by strategy combination, as shown in Table II.

TABLE II. STRATEGY COMBINATION

Operating pattern	Operating pattern	The strategy of first set	The strategy of second set	Final strategy
Master and Slave	The first set is master	A_1	A_2	A_1
	The second set is master	A_1	A_2	A_2
Juxtaposition	Series	A_1	A_2	$A_1 \cap A_2$
	Parallel	A_1	A_2	$A_1 \cup A_2$

For the master and slave operation pattern, the master device strategy is the final strategy. Only when the master device fails, the measures of the auxiliary device strategy are adopted as the final strategy. For the parallel pattern, the union of the two device strategies is the final strategy. For the series pattern of parallel operation, the intersection of the two device strategies is the final strategy.

C. Coordination between Communication System and System Protection Emergency Control Strategy

With the development of communication system, there are not only some stability adjustment measures in the power system but also similar processes in the communication system, as shown in Fig. 2, which are arranged according to the time sequence of different adjustment measures in the communication system. The reason for studying the adjustment measures of the communication system is that, on the one hand, under the background of the current cyber-physical power system, the adjustment of the communication system can effectively reduce the operation risk of the power system; on the other hand, the adjustment of the communication system is simpler and cheaper than changing the load shedding strategy of the power system. Therefore, it is hoped to achieve effective cooperation adjustment between the system protection emergency control strategy and communication system [3].

The existing communication systems often take the following measures: the communication system reasonably allocates routing paths and balances the traffic of different communication links according to the different importance of communication services and the traffic restrictions of each link to minimize the risk of important services and the standard deviation of different link traffic. Generally, for important communication services, the communication system will be equipped with backup routes. When the communication service is interrupted, it can be quickly switched to the standby route, such as the emergency switching operation in the SDH network [9]. In the traditional network based on SDH technology, the direction of information service can be changed directly through the SDH network bridge switching operation, but it will increase the service transmission time, and at the same time, there will be some data loss in the bridge switching process. At present, the OTN network can directly adjust the route at the control layer, which is very convenient for coordinated adjustment with the system protection emergency control strategy.

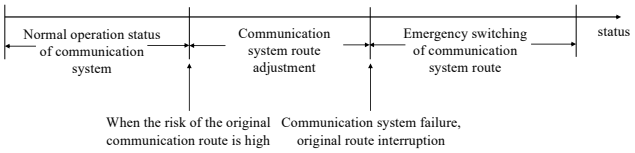


Fig. 2. Communication system adjustment measures.

III. COORDINATED ADJUSTMENT PROCESS CONSIDERING SYSTEM PROTECTION FAILURE RISK

A. Coordinated Adjustment Process of Emergency Control Strategy and Communication System Considering System Protection Failure Risk

Aiming at the possible faults of system protection and the cooperative relationship between emergency control strategy and communication system routing adjustment, this section proposes a cooperative adjustment algorithm. Firstly, the matching check between the expected fault scenarios of power system and the typical fault scenarios of system protection is carried out to judge whether the risk caused by system protection is acceptable. If it is unacceptable, find an alternative route through the communication system first. If alternative routes cannot be obtained, the emergency control strategy of power system shall be adjusted to reduce the failure risk of system protection.

The risk of system protection fault is measured by the consequence caused by the system protection fault. Assuming no fault occurs, the load shedding amount of system protection is $C_{normal-y}$, after the system protection fails, the original strategy may not be implemented effectively. In order to maintain stability, the power system may need to cut off more loads, and the corresponding total load shedding amount is $C_{fault-y}$, the consequence is: $C_y = C_{fault-y} - C_{normal-y}$, C_y represents the consequence caused by the $y-th$ system protection fault [10].

B. The Mathematical Model of the Coordinated Adjustment Algorithm

This section will introduce the mathematical model of the coordinated adjustment algorithm. The coordinated adjustment algorithm proposed in this paper is a sequential decision algorithm.

1) Communication system routing adjustment

Suppose a communication system node connection diagram abstracted from a system protection is shown in Fig. 3 below:

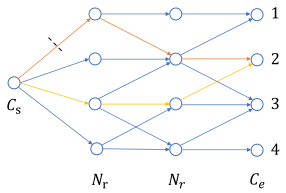


Fig. 3. Communication system node connection diagram.

Problem description: Under the typical fault scenario of system protection (assuming $C_s - N_{r-1}$ link failure), find a new communication route. The source node is C_S and the target node is C_{e-2}. This problem is a path optimization problem. Its objective function and constraints are as follows.

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$$min \sum_{mn} x_{mn}^{k} R_{mn}^{k} \quad (2)$$

$$\sum_{n} x_{mn}^{k} - \sum_{n} x_{nm}^{k} = 1(m \epsilon C_S) \quad (3)$$

$$\sum_{n} x_{mn}^{k} - \sum_{n} x_{nm}^{k} = 0(m \epsilon N_r) \quad (4)$$

$$\sum_{n} x_{mn}^{k} - \sum_{n} x_{nm}^{k} = -1(m \epsilon C_e) \quad (5)$$

$$x_{mn}^{k} + x_{nm}^{k} \leq 1 \quad (6)$$

The above path optimization problem is 0-1 integer programming; C_s is the master station or dispatching center; C_e is the execution station; N_r represents the control substation or intermediate communication equipment; x_{mn}^{k} indicates whether the routes of different services k pass through the mn link; R_{mn}^{k} indicates the failure risk of different links. The objective function indicates that the expected route has the lowest failure risk, mn represents the communication link and k represents the communication service. Equations (3), (4) and (5) respectively represent the flow constraints of nodes.

During normal operation, the routing policy of the system is shown as an orange line. Under the typical fault scenario of $C_s - N_{r-1}$ link, the system can find an available route through optimization, as shown in the yellow line, indicating that the system protection can implement the original emergency control strategy through rerouting.

2) Emergency control strategy adjustment

When the adjustment of the communication system is invalid, the system protection emergency control strategy must be adjusted. The following first introduces two risk indicators, which are represented by the difference between load shedding.

a) Two risk indicators

Undercut risk L_{under}: the impact caused by the failure of the system protection and the failure of the adjusted emergency control strategy to effectively implement.

$$L_{under} = L_{new-i} - L_{new-xin-i} \quad (7)$$

Overcut risk L_{over}: the system protection has not failed, and the adjusted emergency control strategy has an overcut impact compared with the original optimal emergency control strategy.

$$L_{over} = L_{new-xin-i} - L_{old-i} \quad (8)$$

L_{new-i} and L_{old-i} respectively represents the new emergency control strategy under the system protection fault state and the original emergency control strategy; $L_{new-xin-i}$ is the emergency control strategy calculated by the algorithm in this paper.

b) Mathematical model

If the fault of system protection is regarded as a deterministic fault, the redefined emergency control strategy may cause more over switching than the original optimal strategy when there is no fault. If the system protection is considered to be completely reliable, it will cause a greater failure risk to the power system after its failure. Therefore, the weighted sum of undercut risk and overcut risk defined above is taken as the final objective function.

Objective function:

$$\begin{aligned} \min (a \cdot \sum_{m} \sum_{i \in N_G} |\Delta P_{Gi}^{m}| + b \cdot \\ \sum_{m} \sum_{i \in N_S} |L_{new-xin-i}^{m} - L_{new-i}^{m}| + b \cdot \\ \sum_{m} \sum_{i \in N_S} |L_{new-xin-i}^{m} - L_{old-i}^{m}|) \end{aligned} \quad (9)$$

where: a and b respectively represent generator power adjustment and load shedding cost; m represents the number of different power system fault scenarios; N_S represents the number of load nodes.

The DC power flow is used as constraint, other constraints include power angle θ^m, line power flow f_l^m, load shedding $L_{new-xin-i}^{m}$ and generator output adjustment ΔP_{Gi}^{m} constraint.

$$P_{Gi}^{m} + \Delta P_{Gi}^{m} - (P_{Li}^{m} - L_{new-xin-i}^{m}) = B_{fault}^{m} \cdot \theta^m \quad (10)$$

$$f_l^m = \theta_i^m - \theta_j^m / x_{ij} \quad (11)$$

$$f_l^{min} < f_l^m < f_l^{max} \quad (12)$$

$$\theta_i^{min} < \theta_i^m < \theta_i^{max} \quad (13)$$

$$0 \leq L_{new-xin-i}^{m} \leq \varepsilon \quad (14)$$

$$P_{Gi}^{min} \leq P_{Gi}^{m} + \Delta P_{Gi}^{m} \leq P_{Gi}^{max} \quad (15)$$

$$\sum_{i} L_{new-xin-i}^{m} \leq \sum_{i} L_{new-i}^{m} \quad (16)$$

The B_{fault}^{m} and ε represent system admittance matrix and load shedding limit under different faults. The constraint formula (16) ensures that the emergency load shedding amount proposed in this paper does not exceed the load shedding amount when the system protection fails.

The solution process of emergency control strategy L_{new-i}^{m} can be transformed into an optimization problem. The objective function and constraints are as follows. The meaning of this optimization problem is to find the optimal emergency control strategy in the new feasible region after removing the spatial domain associated with the system protection fault equipment from the original controllable strategy space. The variables in objective functions and constraints are consistent with the above meaning, k_i indicates the status of the corresponding system protection equipment. Only some constraints are listed here.

$$min \sum_{i} L_{new-i}^{m} \quad (17)$$

$$P_{Gi}^{m} + \Delta P_{Gi}^{m} - (P_{Li}^{m} + k_i \cdot L_{new-i}^{m}) = B_{fault}^{m} \cdot \theta^m \quad (18)$$

C. Example Results

The coordinated adjustment process in this paper takes the 3-machine 12 node modification system in Fig. 4 as an example. The system protection structure is shown in Fig. 5. MS represents the main station; S represents the sub station; E represents the executive station, and there are 8 executive stations to control different generators and loads respectively; L represents the communication link.

In this example, the system protection adopts the configuration of a single set of devices, so there is no need for a strategy combination.

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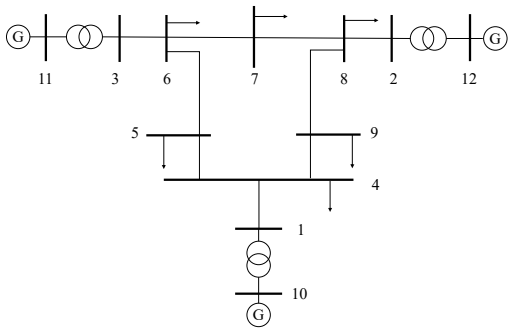


Fig. 4. Modified 3-machine 12 node power system structure.

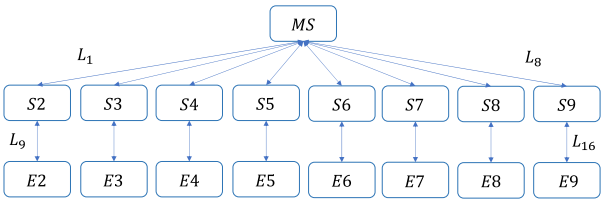


Fig. 5. System protection structure.

1) Typical fault scenarios of system protection

After calculation, the risk to the power system caused by the failure of executive stations 4, 7 and 9 is negligible, almost zero. When other execution stations fail, certain risk losses will be caused to the power system, and a certain amount of load needs to be shed, as shown in Table III. Therefore, according to the definition of typical failure scenarios, the failures of execution stations 5, 6 and 8 are regarded as typical failure scenarios.

When faults of branches 1-4, 3-6 and 2-8 occur in the power system, and faults of executive stations 5, 6 and 8 occur at the same time, the fault risk caused by system protection is unacceptable, and further cooperative adjustment measures need to be taken to reduce the risk.

2) Communication system adjustment

Take the route adjustment process of execution station 5 as an example. When execution station 5 fails, there is no communication route that can be adjusted in the original system, so the emergency control strategy adjustment can only be used to reduce the failure risk.

The above process can also be proved to have no solution by solving the routing optimization problem. Similarly, the fault of other execution stations needs to be adjusted through an emergency control strategy to reduce the risk.

TABLE III. FAILURE SCENARIOS RISK ANALYSIS

System protection fault scenarios	Cumulative risk	Is it a typical fault scenario?
Execution station 5 fault	4.655	yes
Execution station 6 fault	3.6707	yes
Execution station 8 fault	2.9192	yes

3) Emergency control strategy adjustment

When the system protection is normal, the system protection strategy table is shown in Table IV.

TABLE IV. SYSTEM PROTECTION STRATEGY

Power system fault	Generator output adjustment（MW）	Load shedding（MW）
Branch1-4	$\Delta P_{G2}=1.41$ $\Delta P_{G3}=50.59$	$\Delta P_{L5}=-15$
Branch3-6	$\Delta P_{G2}=4$ $\Delta P_{G3}=-85$	$\Delta P_{L5}=-6$ $\Delta P_{L6}=-18.3$
Branch2-8	$\Delta P_{G2}=-163$ $\Delta P_{G3}=71$	$\Delta P_{L8}=-27.64$

When the system protection execution station fails, the alternative emergency control measures are shown in Table V.

TABLE V. NEW STRATEGY IN CASE OF SYSTEM PROTECTION FAILURE

Power system fault	System protection fault	Generator output adjustment（MW）	Load shedding（MW）
Branch1-4	Execution station 5	$\Delta P_{G2}=4$ $\Delta P_{G3}=45.77$	$\Delta P_{L4}=-17.23$
Branch3-6	Execution station 6	$\Delta P_{G2}=4$ $\Delta P_{G3}=-85$	$\Delta P_{L5}=-8.49$ $\Delta P_{L7}=-19.48$
Branch2-8	Execution station 8	$\Delta P_{G2}=-163$ $\Delta P_{G3}=71$	$\Delta P_{L7}=-30.55$

The emergency control strategy calculated by the algorithm in this paper is shown in Table VI.

TABLE VI. EMERGENCY CONTROL STRATEGY BY USING THE OPTIMIZATION METHOD PROPOSED IN THIS PAPER

Power system fault	System protection fault	Generator output adjustment（MW）	Load shedding（MW）
Branch1-4	Execution station 5	$\Delta P_{G1}=-72.3$ $\Delta P_{G2}=3.76$ $\Delta P_{G3}=47.01$	$\Delta P_{L4}=-11.23$ $\Delta P_{L5}=-5$
Branch3-6	Execution station 6	$\Delta P_{G1}=48.72$ $\Delta P_{G2}=4$ $\Delta P_{G3}=-85$	$\Delta P_{L5}=-7.81$ $\Delta P_{L6}=-5$ $\Delta P_{L7}=-14.16$
Branch2-8	Execution station 8	$\Delta P_{G1}=60.7$ $\Delta P_{G2}=-163$ $\Delta P_{G3}=67.45$	$\Delta P_{L7}=-24.55$ $\Delta P_{L8}=-5$

Whether it is the emergency control strategy $L^m_{new-xin}$ calculated by the algorithm proposed in this paper or the original emergency control strategy L^m_{old}, after the system protection fails, it will cause a certain failure risk to the power system. As described above, the fault risk comes from the undercut (the larger the undercut may cause more serious consequences of power system load loss). Here, the fault consequences caused by unit load undercut are set to 100/MW.

It can be seen from the above table that the consequences of undercut of emergency control strategy calculated by using the algorithm proposed in this paper are:

$$c = 18 \times 100 = 1800$$

The emergency control strategy is not adjusted, and the fault consequences of the original emergency control strategy are:

$$c = (17.23 + 21.97 + 30.55) \times 100 = 6975$$

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Through the analysis of the results, it is found that the proposed algorithm can significantly reduce the fault risk caused by the system protection fault. At the same time, it can be seen from the results in Table V and Table VI that the proposed algorithm can also reduce the overcut impact caused by adopting the worst measure---considering the system protection fault as a completely determined fault.

IV. Conclusion

Firstly, the duty strategy of system protection under typical expected fault scenarios is determined. Then, the duty strategy is used to judge whether the impact of the system protection fault on the power system is acceptable. If the impact is unacceptable, the routing is first adjusted to implement the original optimal strategy. If a feasible route cannot be found, a mathematical model is established to adjust the emergency control strategy by comprehensively considering the risk of undercut and overcut. Finally, an example is given to verify the effectiveness of the proposed algorithm, which can effectively reduce the risk caused by system protection failure and improve the reliability of power system.

V. Future Work

In the process of power system emergency control strategy and communication system route adjustment considering the impact of system protection failure, it is currently considered that the failure form of system protection is the failure of communication links between different levels of system protection and other hardware devices. However, in practical engineering applications, there are various fault forms, including communication delay, bit error and other fault forms under the background of large-scale interconnection of power grids. How to adjust the emergency control strategy and communication system when these factors are taken into account can be the next research direction.

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